Fact-Checking Audit & Publication-Venue Suitability Analysis

"Graph-Conditioned Neural Decompilation of Dart AOT Binaries: Pass@k, Reward Adaptation, and Multi-Arm Candidate Ensembling" (Abualazm, AboElHassan, Wassal)

TL;DR

- **The paper's statistics are exceptionally clean** — every arithmetic, McNemar exact-binomial, Wilson-CI, and Holm-Bonferroni value I recomputed matches the manuscript — **but two errors remain that map exactly to the "#1 must-fix" fabricated-reference category from prior rounds**: (a) the claim that MultiPL-E "includes Dart in its polyglot benchmark family" is **FALSE** (MultiPL-E covers 18 languages — none of them Dart), and (b) reference [26] Qwen3 contains **corrupted/fabricated author names** ("Jingren Lin", "Zhenru Cui") that appear nowhere in the real 60-author list.
- **All four frontier models (GPT-5.5, Claude Sonnet 5, DeepSeek V4 Pro, GLM-5.2) and the Azure "gpt-chat-latest" alias are REAL and correctly named as of July 2026** — but the described sampling protocol is technically impossible for Claude Sonnet 5 (its API returns a 400 error when you set non-default temperature/top_p), and "gpt-chat-latest" is an explicitly non-pinned moving snapshot, both reproducibility red flags reviewers will catch.
- **Best venue given the honest "frontier-beats-specialized" negative result: FSE 2027** (CORE A*, paper deadline Oct 2, 2026) as top choice, **USENIX Security 2027 Cycle 2** (A*, Jan 26, 2027) second, **TSE journal** (A*, rolling) third — explicitly avoiding TOSEM because reference [2] is under review there.

Key Findings

Bibliography (Task 1)

- **Two confirmed high-severity errors**: [4] the MultiPL-E-includes-Dart claim (factual) and [26] Qwen3 corrupted author names (fabrication-class).
- **Verified CORRECT** (against arXiv/ACL Anthology/GitHub live sources): [3] SLaDe (Chris Cummins **is** a legitimate author — the flagged concern is unfounded), [7] Idioms, [13] Nova (**genuinely accepted at ICLR 2025**), [17] CodeInverter, [19] SimKO, [21] DeepSeekMath/GRPO, [22] LLM4Decompile, [23] Decompile-Bench, [24] B(l)utter, [27] Yue et al. (**the duplicate "Yang Yue" is REAL** — two different people, 乐洋 and 乐阳, sharing an English name; NOT a duplication error).
- **Not independently re-verified in this audit** (search budget exhausted — recommend a final DBLP pass): [1][2] self-citations, [5][6][8][9][10][11][12][14][15][16][18][20][25].

External claims (Task 2)

- Qwen3-8B-Base exists (`Qwen/Qwen3-8B-Base`, 8.2B params, 36 layers, 32,768 context).

GraphCodeBERT's title confirms data-flow-based representation pretraining.

- **MultiPL-E does NOT include Dart.**
- The HumanEval pass@k estimator $1 - C(n-c,k)/C(n,k)$ is exactly the Codex-paper unbiased estimator — correct.
- Library version strings could not be verified (flag; author should paste an exact `pip freeze`).

Internal consistency (Task 3)

- **All numeric checks pass** — no arithmetic or statistical error found, including the Holm-Bonferroni value the task worried about. Five genuine but minor prose/consistency items need fixing.

Venues (Task 4)

- Open A* deadlines after July 10, 2026: **FSE 2027 (Oct 2)**, ICLR 2027 (Sep 24), AAAI 2027 (Jul 28), **USENIX Sec 2027** C1 (Aug 25) / C2 (Jan 26, 2027), NDSS 2027 fall (Aug 19), IEEE S&P 2027 C2 (Nov 17). ICSE 2027, ASE 2026, ISSTA 2026, EMNLP 2026, NeurIPS 2026, and CCS 2026 are all **closed** for their current edition.

Details

TASK 1 — Reference-by-reference verification

Ref	Verdict	Notes
[1] Abualazm & AboElHassan, *LLMs as Idiomatic Decompilers*, SANER 2026 ERA, arXiv:2604.02278	**UNVERIFIED**	Could not confirm the arXiv ID or SANER 2026 ERA acceptance in this audit. Self-citation — author must confirm 2604.02278 resolves and de-anonymize appropriately.
[2] Abualazm, AboElHassan, Wassal, *Evaluating fine-tuning and metrics…*, arXiv:2607.06125, under review TOSEM	**UNVERIFIED**	Could not confirm arXiv 2607.06125. **TOSEM-overlap concern** — do not submit the present paper to TOSEM while [2] is under review there.
[3] SLaDe, CGO 2024, pp. 67–80	**CORRECT**	Armengol-Estapé, Woodruff, **Chris Cummins**, O'Boyle (Univ. Edinburgh + Meta AI Research), Conference-publishing University of Edinburgh Re… arXiv:2305.12520. Cummins **is** a legitimate author — no discrepancy; the flagged worry is unfounded.

Ref	Verdict	Notes
[4] MultiPL-E, IEEE TSE 49(7):3675–3691, 2023	**AUTHORS CORRECT / CLAIM WRONG (high severity)**	Author list (Cassano, Gouwar, D. Nguyen, S. Nguyen, Phipps-Costin, Pinckney, Yee, Zi, Anderson, Feldman, Guha, Greenberg, Jangda) verified. **But the paper's claim that MultiPL-E includes "Dart in its polyglot benchmark family" is FALSE.** The 18 MultiPL-E target languages are C++, C#, D, Go, Java, JavaScript, Julia, Lua, PHP, Perl, R, Racket, Ruby, Rust, Scala, Swift, TypeScript, and Bash/Shell (Readthedocs) (per the MultiPL-E docs and Cassano et al.: "translate … to 18 other programming languages"). (arxiv) (arxiv) **Dart is absent.** Note the easy-to-miss confusion: MultiPL-E includes **D** (a different language), not Dart. Delete or correct this claim.
[5] Chen et al., *Evaluating LLMs trained on code* (Codex), arXiv:2107.03374	**LIKELY CORRECT**	Suchir Balaji, Jan Leike, Dario Amodei, Ilya Sutskever, Wojciech Zaremba are all genuine Codex authors; consistent with corroborating citations. Recommend a final DBLP check of exact ordering/length.
[6] DIRTY, USENIX Security 2022	**NOT RE-VERIFIED**	Well-established (Chen, Lacomis, Schwartz, Le Goues, Neubig, Vasilescu); confirm on DBLP.
[7] Idioms, arXiv:2502.04536	**CORRECT**	Luke Dramko, Claire Le Goues, Edward J. Schwartz (CMU). Title "…Joint Code and Type Definition Prediction" matches. (arxiv) (arXiv)
[8] Falliere, JEB Dart AOT snapshot helper plugin, PNF Software blog, 2022	**NOT RE-VERIFIED**	Could not confirm the specific blog post in this audit; author should re-check the URL/date.
[9] GraphCodeBERT, ICLR 2021	**LIKELY CORRECT**	Guo et al.; title confirms data-flow pretraining. Confirm the full author list on DBLP.
[10] Hosseini & Dolan-Gavitt, *Beyond the C*, BAR 2022	**LIKELY CORRECT**	Appears in multiple corroborating reference lists.
[11] LoRA, ICLR 2022	**NOT RE-VERIFIED**	Hu et al.; well-established.
[12] reFlutter (Impact-I / PT SWARM), GitHub 2021	**NOT RE-VERIFIED**	Confirm the repo attribution.

Ref	Verdict	Notes
[13] Nova, ICLR 2025	**CORRECT**	Nan Jiang, Chengxiao Wang, Kevin Liu, Xiangzhe Xu, Lin Tan, Xiangyu Zhang, Petr Babkin. Semantic Scholar **Genuinely accepted at ICLR 2025** (poster; ICLR arXiv:2311.13721) — venue claim is correct.
[14] Katz, Ruchti, Schulte, SANER 2018	**NOT RE-VERIFIED**	Well-established.
[15] DIRE, ASE 2019	**NOT RE-VERIFIED**	Lacomis et al.; well-established.
[16] AlphaCode, Science 378(6624):1092–1097, 2022	**LIKELY CORRECT**	Li et al.; the filter-and-cluster candidate-selection claim is a genuine AlphaCode feature.
[17] CodeInverter, arXiv:2503.07215	**CORRECT**	Peipei Liu, Jian Sun, Rongkang Sun, Li Chen, Zhaoteng Yan, Peizheng Zhang, Dapeng Sun, Dawei Wang, Xiaoling Zhang, Dan Li. arxiv The CFG + data-mapping injection into LLM decompilation claim is verified. arXiv
[18] SimPO, NeurIPS 2024	**NOT RE-VERIFIED**	Meng, Xia, Chen; well-established.
[19] SimKO, arXiv:2510.14807	**CORRECT**	Ruotian Peng, Yi Ren, Zhouliang Yu, Weiyang Liu, Yandong Wen. GitHub The "RLVR over-concentration on top-1 candidates suppresses pass@K" claim is verified. arXiv (Later arXiv versions retitle to "Beyond the Sampled Token"; arXiv citing the v1/v2 title is fine.)
[20] CodeBLEU, arXiv 2020	**NOT RE-VERIFIED**	Ren et al.; well-established.
[21] DeepSeekMath, arXiv:2402.03300	**CORRECT**	Shao et al.; **GRPO is introduced here** as a critic-free, group-relative variant of PPO ("foregoes the critic model, instead estimating the baseline from group scores") arXiv — verified.
[22] LLM4Decompile, EMNLP 2024, pp. 3473–3487	**CORRECT**	Hanzhuo Tan, Qi Luo, Jing Li, Yuqun Zhang (4 authors). ACL Anthology 2024.emnlp-main.203; page range matches. ACL Anthology
[23] Decompile-Bench, arXiv:2505.12668	**CORRECT**	Tan, Tian, Qi, Liu, Gao, Wang, Luo, Li, Zhang arXiv — matches the paper's list.

Ref	Verdict	Notes
[24] B(l)utter, GitHub 2023	**CORRECT**	Worawit Wangwarunyoo, `worawit/blutter`. GitHub Confirmed: compiles the Dart AOT runtime and extracts information from `libapp.so` GitHub — i.e., snapshot introspection / metadata recovery, **not** source-level, test-executed neural decompilation. The paper's characterization is accurate.
[25] ReSym, CCS 2024	**NOT RE-VERIFIED**	Xie et al.; well-established.
[26] Qwen3 Technical Report, arXiv:2505.09388	**FABRICATED NAMES (high severity)**	The real 60-author list reads "…**Jingren Zhou**, **Junyang Lin**, Kai Dang… jsDelivr **Zeyu Cui**, **Zhenru Zhang**, Zhipeng Zhou, Zihan Qiu." arXiv The paper's "**Jingren Lin**" conflates *Jingren Zhou* + *Junyang Lin*; "**Zhenru Cui**" conflates *Zeyu Cui* + *Zhenru Zhang*. Neither invented name appears anywhere in the actual list — this is exactly the fabricated-author-list problem flagged as #1 in prior rounds. Fix to the real names (or cite as "Qwen Team" / "An Yang et al.").
[27] Yue et al., arXiv:2504.13837	**CORRECT**	Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, Gao Huang. **The duplicate "Yang Yue" is genuine** — the paper's own footnote notes two different authors (乐洋 and 乐阳) share the English name; arXiv NOT a duplication error. Discusses pass@1 vs. large-k pass@k — verified. NeurIPS 2025 poster.

TASK 2 — External factual claims

Frontier models — all REAL and correctly named as of July 2026:

- **GPT-5.5**: OpenAI, released April 23, 2026 (ChatGPT/Codex), API April 24, 2026. Real. Wikipedia Wikipedia
- **Claude Sonnet 5**: Anthropic, released June 30, 2026, API ID `claude-sonnet-5`. Real. Anthropic
- **DeepSeek V4 Pro**: `deepseek-ai/DeepSeek-V4-Pro` (1.6T total / 49B active), preview launched April 24, 2026; available on OpenRouter. Real. Hugging Face WaveSpeedAI
- **GLM-5.2**: Zhipu AI / Z.ai, released June 13, 2026 (~744–753B MoE); available on OpenRouter. Real. Fello AI
- **gpt-chat-latest**: a genuine **Microsoft Foundry** product name for the GPT-5.5 Instant release (OpenAI API name `chat-latest`). Real. Microsoft Learn

Non-pinned snapshot (reproducibility red flag). OpenAI's API changelog states `chat-latest` "points to the latest Instant model currently used in ChatGPT ... The underlying model snapshot will be regularly updated" — a moving target. OpenAI's own guidance is verbatim: "We recommend leveraging GPT-5.5 for production API usage, but feel free to use this model to test the latest improvements for chat use cases." Using `gpt-chat-latest` for the headline pass@10=0.7013 result undermines reproducibility; pin to a dated GPT-5.5 snapshot. Openai Openai

Timing. GPT-5.6 (Sol/Terra/Luna) reached general availability July 9, 2026 — one day before the research date — so GPT-5.5 is already one generation old. Not an error, but note the fast-moving landscape in the paper. Coursiv Blog

Other claims: Qwen3-8B-Base checkpoint exists as claimed; the HumanEval pass@k estimator is the exact Codex-paper form; GraphCodeBERT, DeepSeekMath/GRPO, CodeInverter, SimKO, AlphaCode, and Yue et al. related-work characterizations are all accurate (see Task 1). **MultiPL-E does not include Dart** (critical — see [4]).

Library versions — UNVERIFIED. Search budget was exhausted before PyPI could be checked, and the version-history pages were not fetchable. transformers was at 4.51+ in mid-2025, so a 5.9.0 by mid-2026 implies a 5.x major series — plausible but unconfirmed; torch 2.11.0+cu128, peft 0.18.0, torch-geometric 2.7.0, accelerate 1.6.0, and Python 3.12.7 are all plausible but unverified. Author should paste an exact `pip freeze` and confirm none are anachronistic.

TASK 3 — Internal consistency (all recomputed by hand)

Coverage fractions — ALL CORRECT: 43/154=0.2792; 110/154=0.7143; 108/154=0.7013; 112/154=0.7273; 78/154=0.5065; 109/154=0.7078; 24/154=0.1558; 28/154=0.1818; 26/154=0.1688; 17/154=0.1104; 8/154=0.0519.

McNemar exact two-sided binomial p-values — ALL CORRECT: (3,5)→0.727; (3,1)→0.625; (5,37)→4.43e-7; (2,3)→1.000; (1,2)→1.000; (7,58)→4.3e-11; (19,31)→0.119; (9,52)→1.8e-8; (4,83)→3.0e-20; (7,3)→0.344; (7,5)→0.774; (5,12)→0.143; (2,18)→4.0e-4; (0,24)→1.2e-7; (19,0)→3.8e-6.

Discordant-pair ↔ compile-task-count consistency — ALL CORRECT: GRPO 84−58+7=33; RS-SFT 84−41+11=54; CFG-only 84−31+19=72; text-only 84−52+9=41; p128 84−83+4=5.

Pass@10 task-count consistency — ALL CORRECT: GRPO 24−3+7=28; RS-SFT 24−5+7=26; CFG-only 24−12+5=17; text-only 24−18+2=8; Union 24+19=43.

Wilson 95% CIs — ALL CORRECT: 84/126→[0.5805,0.7430]; 33/126→[0.1930,0.3449]; 24/154→[0.1070,0.2214]; 28/154→[0.1289,0.2502]; 43/154→[0.2144,0.3548]; 0/154→[0.0000,0.0243].

Holm-Bonferroni (m=11) — ALL CORRECT and procedurally valid. Sorted ascending: 3.0e-20 (p128 compile) ×11 = 3.3e-19; **4.3e-11 (GRPO compile) ×10 = 4.3e-10** — the task's concern is unfounded: ×10 is exactly right for rank 2 under m=11 (it is not a rank-1/m=10 error); 1.8e-8 ×9; 1.2e-7 (p128 pass) ×8 = 9.5e-7; 3.8e-6 (union) ×7 = 2.7e-5; 3.6e-5 (RS-SFT compile) ×6 = 2.2e-4; 4.0e-4 (text-only pass) ×5 = 2.0e-3; 0.119 ×4; 0.143 ×3; 0.344 (GRPO pass) ×2 = 0.688; 0.774 ×1. All claimed adjusted values are consistent.

Table 5 — ALL CORRECT: Sel.⩽Oracle and Orig.⩽Oracle hold in every row (0.0325⩽0.0519; 0.0584⩽0.0779; 0.0844⩽0.1104; 0.0844⩽0.1364).

Ratio claims — ALL CORRECT: "roughly four times" 0.7013/0.1818=3.86 ✓; "more than twice the union" 0.7013/0.2792=2.51 ✓; "threefold" G3 vs text-only 0.1558/0.0519=3.00 ✓.

Monotonicity / nesting — ALL CORRECT: Table 1's 14 pass@10 values are monotonically non-increasing; v2 compile@10 (0.9870/0.9675/0.9805) ⩾ pass@10 (0.7078/0.7013/0.7013); compile@10 0.955 ≈ Table 2's 0.9545; 43+111=154.

Five genuine minor items to FIX (none are computational errors):

1. The artifact section says "six GPT-5.5 ablation pools" but Table 3 has **seven** rows (4 v1 + 3 v2) — correct the count.
2. "Three further frontier models also exceed the union" is **fragile**: GLM-5.2 at 0.2857 beats the union's 0.2792 by only 0.0065 (~1 task). Soften to "narrowly exceed" or state the margin.
3. The "**G1**" label for the text-only arm appears without a definition — add it to the Appendix A glossary or drop the label.
4. The ablation full-assembly row (0.7143) vs. the main run (0.7013) differ by ~2 tasks, attributed to different sampling settings; presenting both **unreconciled** invites a "re-run under matched settings" demand — add a one-line reconciliation.
5. "46/154 records show lower prompt-token accounting under the cap" implies only 46 tasks exceeded ~2,048 tokens, which sits awkwardly against the "capability cliff near 200 instructions" framing from prior work — clarify the interpretation.

TASK 4 — Venue suitability (CORE2023 ranks; deadlines after July 10, 2026)

Open, viable venues:

- **FSE 2027 — CORE A*.** Paper deadline **Oct 2, 2026**; Shenzhen, July 2027. Best topical + cultural fit: FSE explicitly welcomes rigorous empirical and negative-results SE papers, and LLM-for-code/decompilation is squarely in scope. ~3 months to fix artifacts.
- **USENIX Security 2027 — CORE A*.** Cycle 1 **Aug 25, 2026** (registration Aug 18); Cycle 2 **Jan 26, 2027** (abstract Jan 19); Denver, Aug 2027. Strong fit — decompilation/reverse-

engineering heritage (DIRTY appeared at USENIX Security); multi-cycle model tolerates revision.

- **IEEE S&P 2027 — CORE A*.** Cycle 2 paper **Nov 17, 2026** (abstract Nov 10); ~May 2027, Montreal. (Cycle 1, Jun 11, 2026, is closed.)
- **NDSS 2027 — CORE A*.** Fall cycle **Aug 19, 2026**; Seoul, Mar 2027.
- **ICLR 2027 — CORE A*.** Paper **Sep 24, 2026** (abstract Sep 19). Hard sell — "specialized fine-tuning loses to frontier zero-shot" is a tough narrative at ML venues, and the small single-language (154-task Dart) benchmark with no seed averaging is a reviewability risk against the A* bar.
- **AAAI 2027 — CORE A*.** Paper **Jul 28, 2026** (abstract Jul 21). Very tight; same negative-result concern as ICLR.
- **SANER 2027 — CORE A.** Abstract **Sep 21, 2026**; Richmond, VA, Mar 2027. Excellent topical fit, but it is CORE A (below the A* target) and the author's prior work [1] is already at SANER 2026.
- **CCS 2027 — CORE A*** (~Jan 2027, unannounced); **ISSTA 2027 — CORE A** (~late Jan 2027, unannounced); **ICSME 2027 — CORE A** (~Feb/Mar 2027, unannounced).
- **Journals: TSE (A*)** and **TOSEM (A*)** rolling; **EMSE (A)** rolling. **TOSEM carries a conflict** — reference [2] is under review there.

Closed for the current edition (deadline before July 10, 2026): ICSE 2027 (single cycle, Jun 30), ASE 2026 (Mar 26), ISSTA 2026 (Jan 29), ICSME 2026 (Feb 27), EMNLP 2026 (ARR May 25), NeurIPS 2026 (May 6), CCS 2026 (both cycles), IEEE S&P 2027 Cycle 1 (Jun 11), NDSS 2027 summer, ACSAC 2026 main track (May 26).

Ranked top 3 recommendations:

1. *FSE 2027 (A, Oct 2, 2026).** The best home for an honest empirical/negative result; the deadline gives roughly three months to close artifact gaps and fix the two fabrication-class items. This supersedes the earlier EMNLP-2026 suggestion (now closed) and is stronger than FSE 2027's SE-track siblings for a negative empirical result.
2. *USENIX Security 2027 Cycle 2 (A, Jan 26, 2027).** Values reverse engineering; the two-cycle model tolerates revise-and-resubmit; more runway to add a second language or seed averaging if reviewers demand it.
3. *TSE journal (A, rolling).** No page pressure for the heavy statistics, robust to the "small benchmark" objection via extended empirical framing, and it sidesteps the TOSEM overlap. Slower turnaround is the trade-off.

Expected major reviewer objections across all venues: (i) 154-task, single-language (Dart) benchmark is small; (ii) single-run point estimates with no seed averaging; (iii) missing checkpoint weights/SHA-256 hashes and reconstructed Table-5 run commands (artifact-evaluation risk); (iv) mismatched/undocumented per-model sampling — especially the Claude Sonnet 5 impossibility; (v) frontier baselines are moving targets (`gpt-chat-latest` unpinned; GPT-5.6 already GA).

Recommendations

1. **Fix the two fabrication-class items first.** Correct the Qwen3 [26] author list to the real names (Jingren Zhou, Junyang Lin, Zeyu Cui, Zhenru Zhang — or cite as "Qwen Team"), and **delete the false "MultiPL-E includes Dart" claim** in [4]/intro (note the D-vs-Dart trap). Then run every reference through DBLP once more, prioritizing the unverified set [1][2][5][6][8][9][10][11][12][14][15][16][18][20][25].
2. **Confirm the self-citation arXiv IDs** (2604.02278, 2607.06125) actually resolve. If [2] remains under review at TOSEM, do **not** target TOSEM for this paper.
3. **Resolve the sampling contradiction.** Either re-run the frontier comparison under each model's actually-supported settings (Claude Sonnet 5 forbids non-default temperature/top_p and returns a 400 error), or explicitly document per-model default sampling and drop the uniform temp-0.7/top_p-0.95 claim. **Pin GPT-5.5 to a dated snapshot**, not `gpt-chat-latest`, and disclose it. Reconcile or matched-re-run the 0.7143 (ablation) vs. 0.7013 (main) full-assembly gap.
4. **Close the artifact gaps** (publish checkpoint SHA-256 hashes and real, executed run commands for the Table-5 arms) before any A*/A submission that runs artifact evaluation — this was a prior-round concern and will otherwise fail or delay AE.
5. **Fix the five minor internal items** (six-vs-seven pools; fragile "exceed the union" phrasing; undefined "G1" label; unreconciled 0.7143/0.7013; the 46/154 interpretation).
6. **Submit to FSE 2027 (Oct 2, 2026).** Thresholds that would change this staging: if artifacts (hashes + run commands) cannot be completed and the sampling re-run done by late September 2026, target **USENIX Security 2027 Cycle 2 (Jan 26, 2027)** for the extra runway; if reviewers or your own reading conclude the 154-task single-language benchmark is fatally small for a conference A* bar, pivot to a **TSE journal** submission where an extended empirical treatment is expected.

Caveats

- Library version strings and references [1][2][5][6][8][9][10][11][12][14][15][16][18][20][25] were **not** independently re-verified in this audit (web-search budget was exhausted); treat these as "verify against DBLP/PyPI," not "verified correct." The two confirmed errors ([4] Dart, [26] Qwen3) and the verified-correct set were checked against live primary sources (arXiv, ACL Anthology, IEEE/ACM, GitHub, Hugging Face, vendor docs).
- Some 2027 venue dates (ISSTA, ICSME, CCS, ACL) are not yet officially announced; those are prior-cycle estimates and should be reconfirmed when CfPs post. FSE's CORE rank is authoritatively A* on the official CORE portal despite one aggregator's contradictory "B" scrape.
- Frontier-model facts reflect sources dated through early July 2026; the landscape is moving fast (GPT-5.6 reached GA July 9, 2026, and Anthropic's Fable 5 / Mythos 5 access was under U.S. government restriction during this period).